

Mutual Fund Analytics Capstone Project

1. Introduction

The Mutual Fund Analytics Capstone Project is designed to analyze mutual fund performance, investor behaviour, industry growth trends, and portfolio risk using Python, SQLite, and Power BI. The project integrates multiple datasets and transforms raw financial data into meaningful business insights through data analytics and visualization techniques.

2. Project Objectives

- Build an end-to-end ETL pipeline for mutual fund data.
- Store and manage data using SQLite.
- Perform Exploratory Data Analysis (EDA).
- Calculate advanced performance metrics.
- Develop an interactive Power BI dashboard.
- Generate actionable investment insights.

3. Datasets Used

The project uses the following datasets:

- Fund Master
- NAV History
- AUM by Fund House
- Monthly SIP Inflows
- Category Inflows
- Industry Folio Count
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

4. Technology Stack

- Python
- Pandas
- NumPy
- SQLite

- Matplotlib
- Seaborn
- Plotly
- Power BI

5. ETL Pipeline

Extract

Data was collected from multiple mutual fund datasets.

Transform

- Missing value treatment
- Data type corrections
- Duplicate removal
- Date formatting
- Data validation

Load

Cleaned datasets were loaded into SQLite database tables.

6. Database Design

Tables Created:

- dim_fund
- fact_nav
- fact_aum
- fact_performance
- fact_transactions

The database stores over 46,000 NAV records and 32,000+ transaction records.

7. Exploratory Data Analysis

NAV Trend Analysis

Daily NAV trends were analyzed across all schemes from 2022–2026.

AUM Growth Analysis

SBI Mutual Fund maintained the highest AUM among all fund houses.

SIP Trend Analysis

Monthly SIP inflows showed consistent growth and reached record highs in late 2025.

Category Inflow Analysis

Equity-oriented categories attracted the highest net inflows.

Investor Demographics

The 26–35 age group contributed the largest investment volume.

Geographic Analysis

T30 cities generated significantly higher investment activity compared to B30 cities.

Folio Growth Analysis

Total industry folios nearly doubled during the analysis period.

8. Performance Analytics

Daily Returns

Calculated daily percentage returns for all mutual fund schemes.

CAGR

Computed long-term growth rates using annualized returns.

Sharpe Ratio

Measured risk-adjusted returns using a 6.5% risk-free rate.

Sortino Ratio

Evaluated downside risk-adjusted performance.

Alpha and Beta

Compared scheme returns against benchmark performance.

Maximum Drawdown

Identified the largest decline from historical peaks.

Fund Scorecard

Developed a composite ranking model using:

- 30% Return Rank
- 25% Sharpe Rank
- 20% Alpha Rank
- 15% Expense Ratio Rank
- 10% Drawdown Rank

9. Advanced Analytics

Value at Risk (VaR)

Computed 95% VaR to estimate downside risk.

Cohort Analysis

Analyzed investment behavior by age group and city tier.

Fund Recommendation Engine

Recommended top-performing funds based on 3-year returns.

10. Power BI Dashboard

Page 1 – Executive Dashboard

- Total Funds
- Total AUM
- Average NAV
- Average Sharpe Ratio
- NAV Trend

Page 2 – Performance Analytics

- Risk vs Return
- Alpha vs Beta
- Drawdown Analysis
- Fund Rankings

Page 3 – Investor Analytics

- Age Distribution
- Gender Distribution
- City Tier Analysis
- State-wise Investments

Page 4 – Industry Analytics

- SIP Trends
- Folio Growth
- Category Inflows
- AUM Distribution

11. Key Findings

1. Small Cap funds delivered the highest 3-year returns.
2. SBI Mutual Fund dominated AUM contribution.
3. Investors aged 26–35 formed the largest investor segment.
4. T30 cities generated higher SIP investments.
5. SIP inflows reached record levels during 2025.
6. Top-ranked funds achieved superior Sharpe and Sortino ratios.
7. Small Cap funds dominated the recommendation engine.
8. Industry folio counts nearly doubled from 2022–2025.
9. Risk-adjusted performance varied significantly across categories.
10. Power BI dashboards improved investment decision visibility.

12. Conclusion

The Mutual Fund Analytics Project successfully demonstrated the complete analytics lifecycle from data ingestion to dashboard visualization. The project delivered meaningful insights into fund performance, investor behavior, and industry trends while showcasing practical applications of Python, SQLite, and Power BI in financial analytics.

13. Future Enhancements

- Streamlit Web Application
- Portfolio Optimization
- Monte Carlo Simulation
- Automated Email Reporting
- Real-Time NAV Monitoring